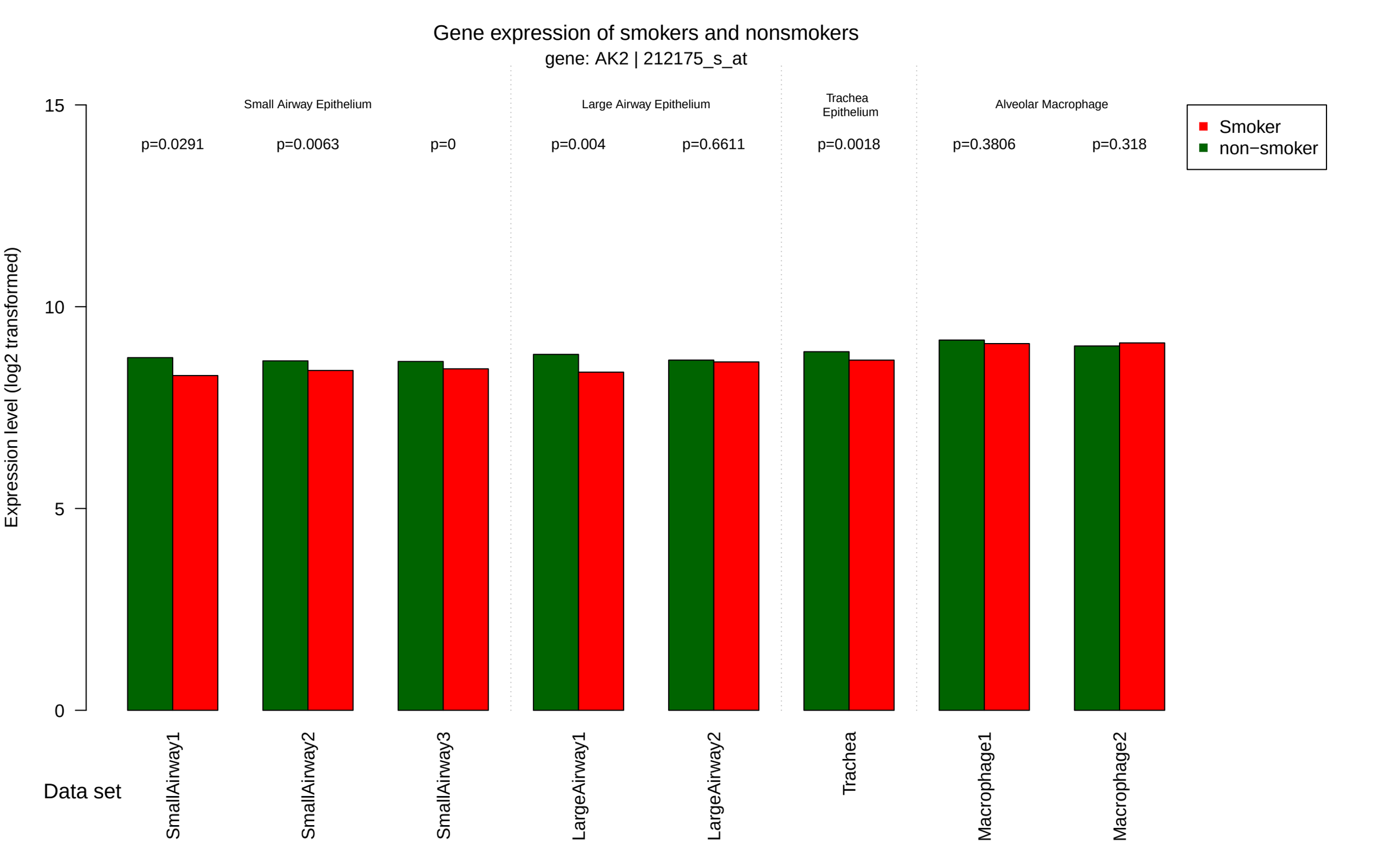
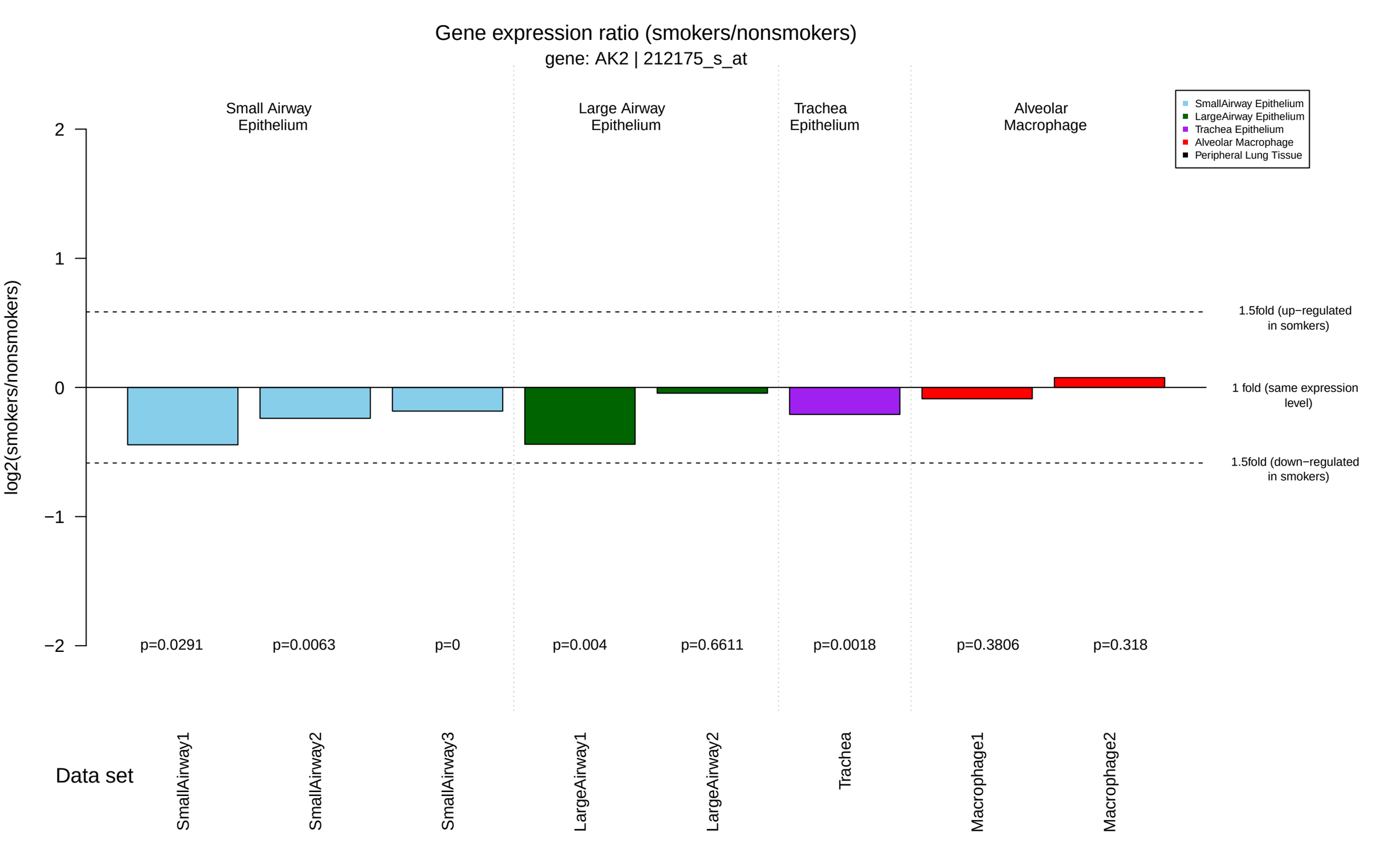
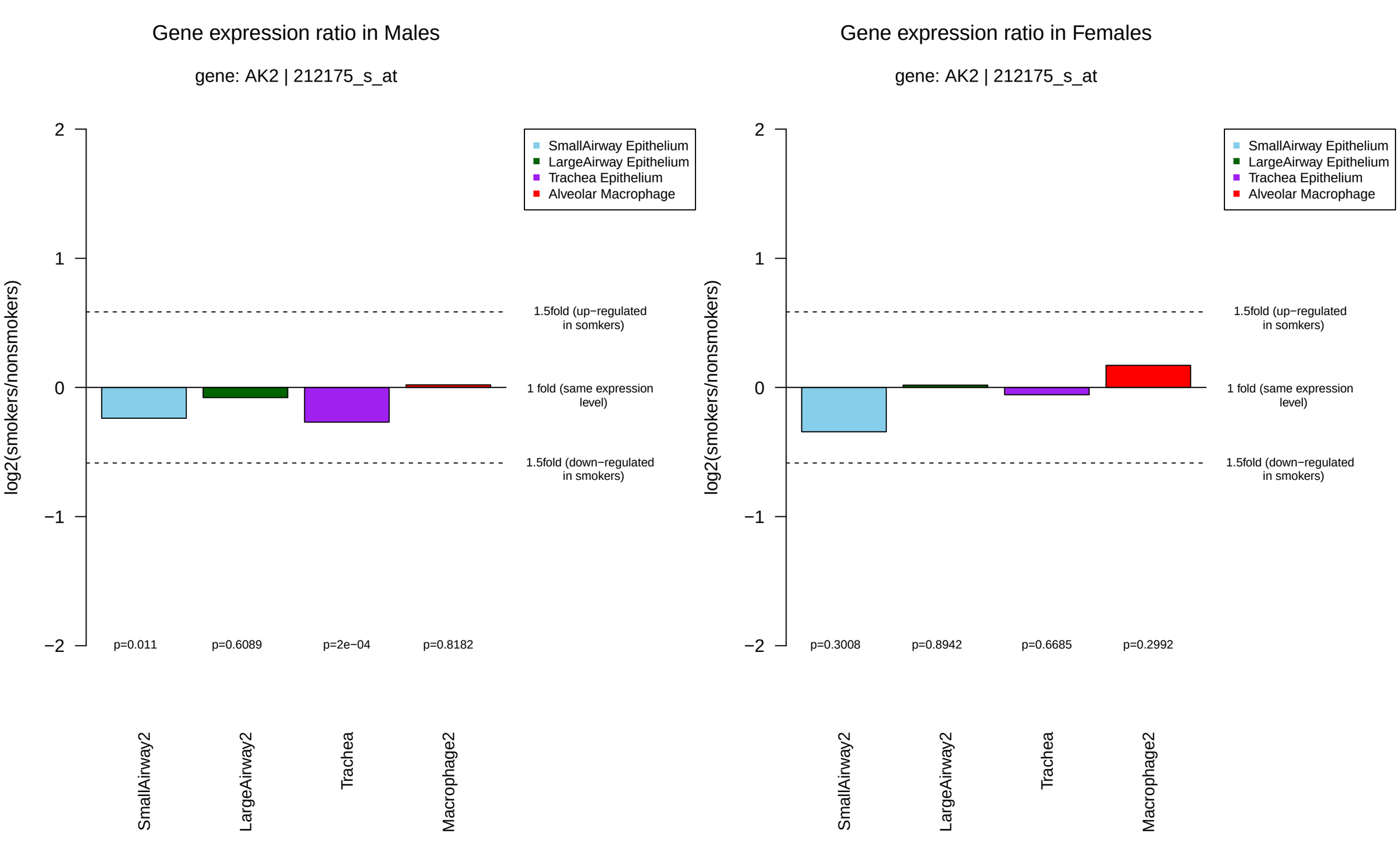
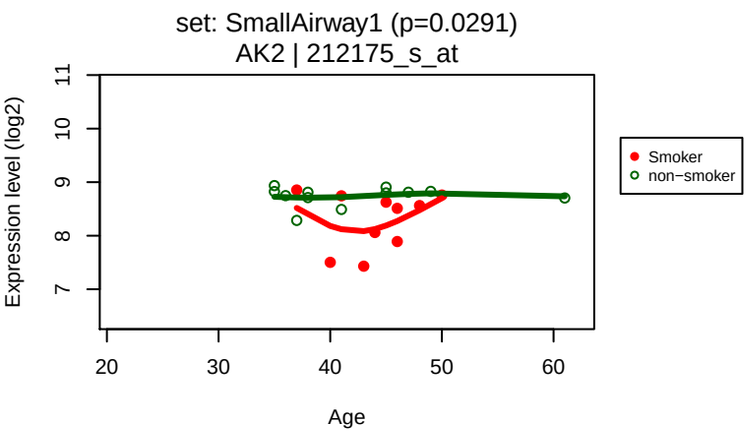
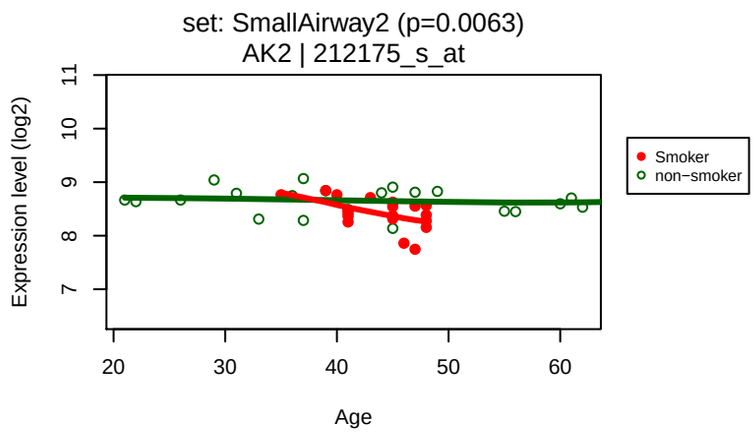
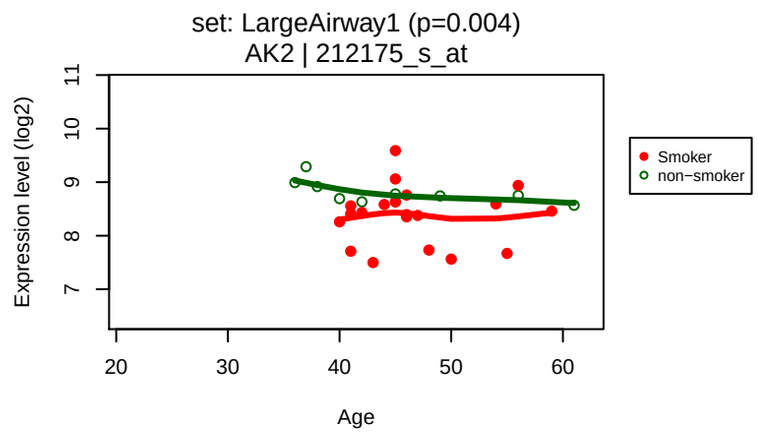

Gene expression of smokers and nonsmokers
gene: AK2 | 212175_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0291
p=0.0063
p=0
p=0.004
p=0.6611
p=0.0018
p=0.3806
p=0.318
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

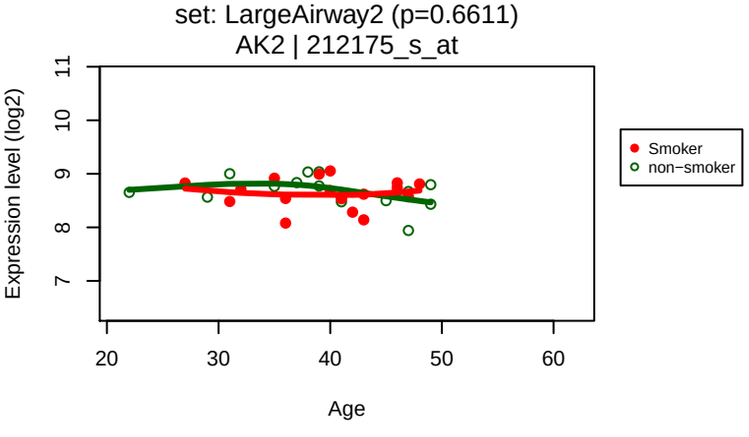

Gene expression ratio (smokers/nonsmokers)
gene: AK2 | 212175_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0291
p=0.0063
p=0
p=0.004
p=0.6611
p=0.0018
p=0.3806
p=0.318
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

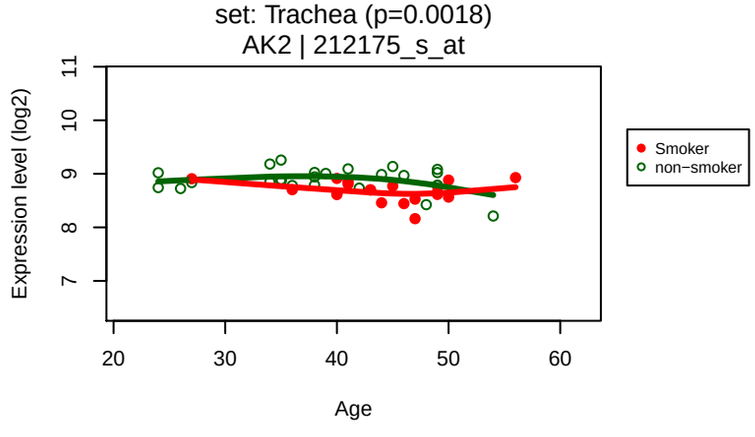

Gene expression ratio in Males
gene: AK2 | 212175_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.011
p=0.6089
p=2e−04
p=0.8182
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AK2 | 212175_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3008
p=0.8942
p=0.6685
p=0.2992
SmallAirway2
LargeAirway2
Trachea
Macrophage2

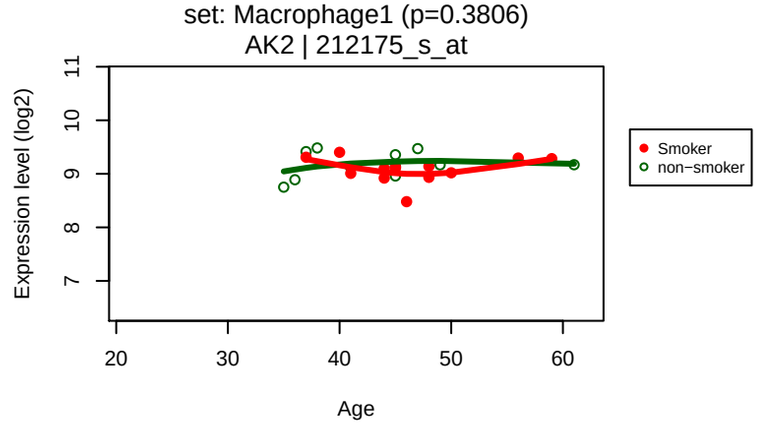

set: SmallAirway1 (p=0.0291)
AK2 | 212175_s_at
Expression level (log2)
Age
Smoker
non−smoker

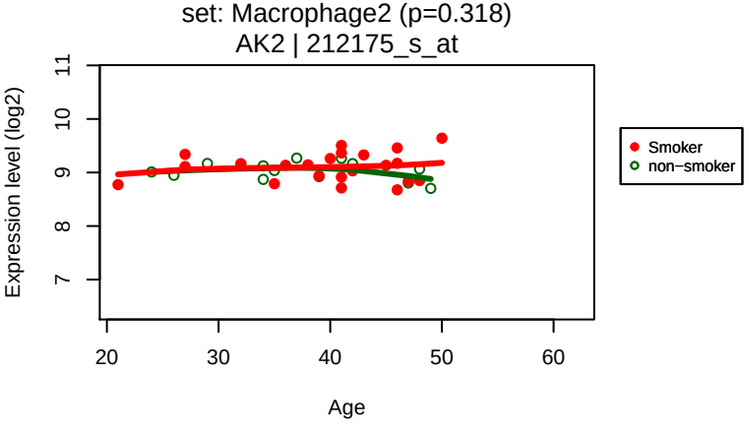

set: SmallAirway2 (p=0.0063)
AK2 | 212175_s_at
Expression level (log2)
Age
Smoker
non−smoker

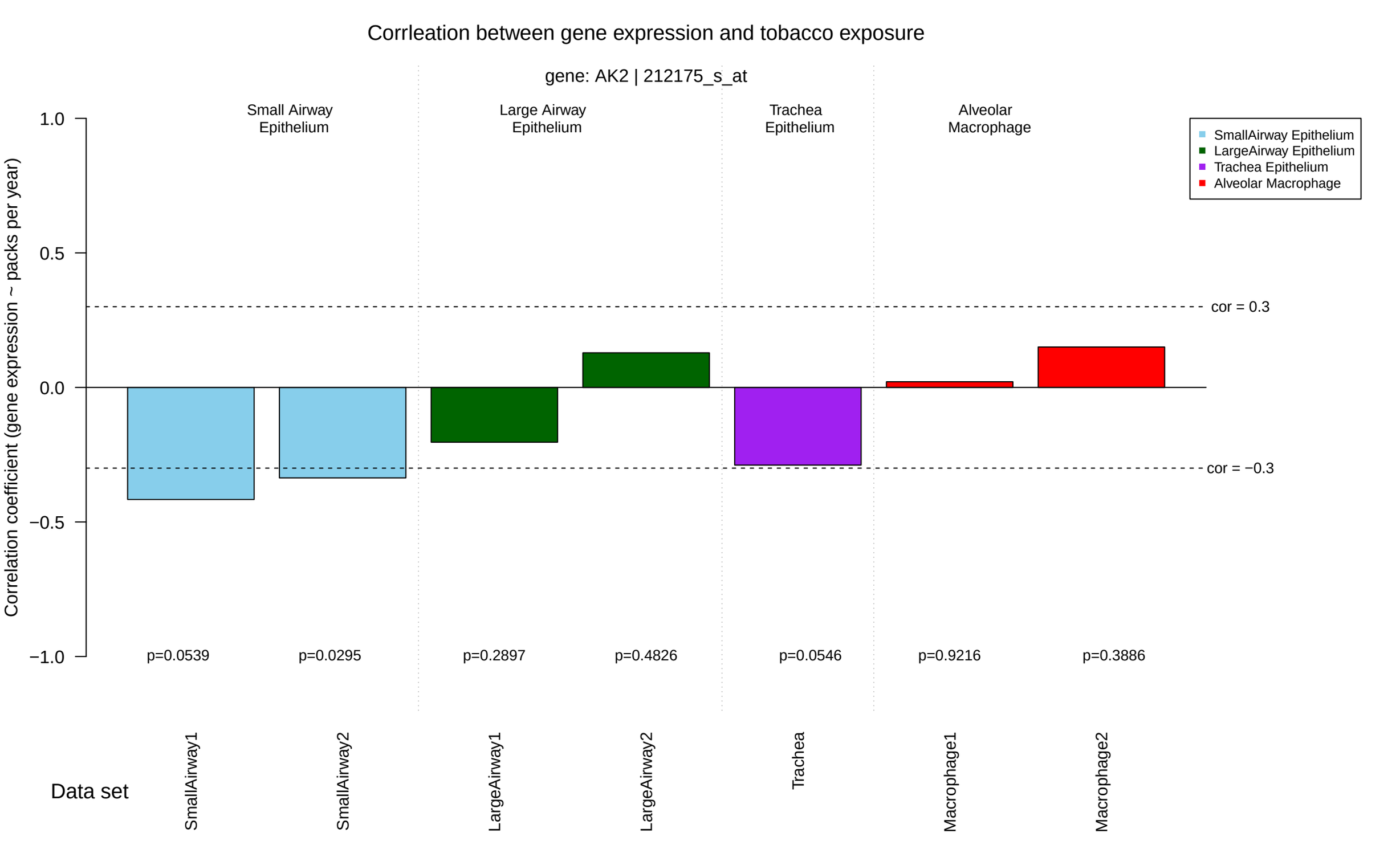

set: LargeAirway1 (p=0.004)
AK2 | 212175_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6611)
AK2 | 212175_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0018)
AK2 | 212175_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3806)
AK2 | 212175_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.318)
AK2 | 212175_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AK2 | 212175_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0539
p=0.0295
p=0.2897
p=0.4826
p=0.0546
p=0.9216
p=0.3886
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2